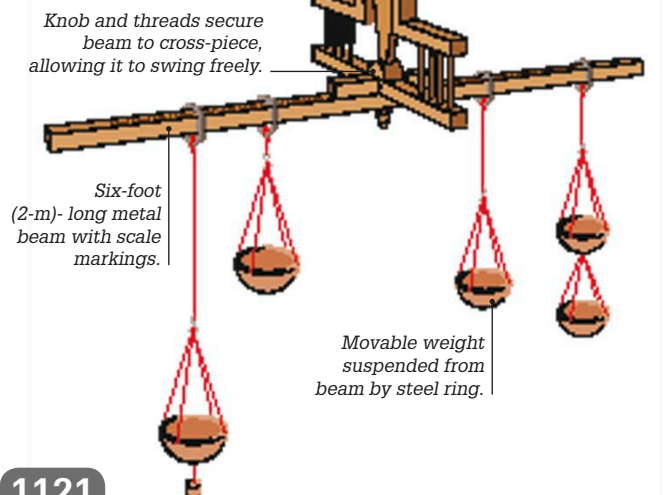


Al-Khazini devised a model of the balancing point of a beam, which depended on weights and their distance from a center point.



1121

Theories of balance and gravity

The Arab scholar al-Khazini published a work on balance and equilibrium. In it he put forward a version of the theory of gravity, stating that the weight of a heavenly body depended on its distance from the center of the Universe.

1120

1145

1126

Key translation

English philosopher Adelard of Bath traveled widely in Italy, Sicily, and the Middle East, becoming familiar with the works of Arab scholars. He translated the famous mathematician al-Khwarizmi's *Astronomical Tables of Sindhind*, spreading knowledge of them to western Europe.

1121

Chinese grid map

A map carved on a stone tablet in Sichuan, China, made the first known use of grid squares to show scale. Known as the *Jiu You Shouling tu*, it has around 1,400 place names and is a sign of how sophisticated Chinese mapmaking had become.

Traveling
the world
See pages
92–93

“Seeing the abundance of books in Arabic... he learned... Arabic... in order to translate them.”

Life of Gerard of Cremona (an Italian scholar), c 12th century

c 1048–1131

OMAR KHAYYAM

Persian poet and philosopher Omar Khayyam was also a talented mathematician and astronomer. By the age of 25, he had written an important work on music and one on algebra. Later, translations of his poetry made him famous in the West, but in the Islamic world of his time, he was famed as a scientist. In 1073, the ruler of Persia, Malik Shah, invited him to set up an observatory in Isfahan. There Omar Khayyam made many important observations and compiled a set of astronomical tables.

Length of a year

While at Isfahan, Omar Khayyam calculated the length of the year to be 365.24219858156 days. This is correct to five decimal places and shows remarkably precise measurement, considering the astronomical instruments available to him.



Book on algebra

In his book on algebra, Omar Khayyam used geometrical methods to solve cubic and quadratic equations. He turned the numbers in the equations into curves and found the solution where they intersected. This technique was very advanced for its time.

